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Remagnetization of Cambrian to Triassic Sedimentary Rocks of the Paegunsan Syncline of the Okch'on Zone, South Korea

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Sixty-eight samples are collected from 24 sites on either side of the wing of the synclinal structure of the Choson and P'yongan Supergroups. Almost all specimens show a stable behavior with respect to the alternating field demagnetization and thermal demagnetization. The sedimentary sequences are normally magnetized and their directions of natural remanent magnetization (NRM) ($D=-2.7°$, $I=58.1°$, $\alpha_{95}=5.2°$) are along the present geocentric axial dipole field in situ. The fold test is negative at 99% probability level. This evidence indicates that the original NRM acquired during the formation of the sedimentary strata has been overcome by the new NRM which was acquired quite recently, probably in the Brunhes epoch. The new NRM is carried by hematite and pyrrhotite. These minerals have been produced in the sedimentary strata due to metamorphism which had occurred before the Brunhes epoch. We conclude that the remagnetization of the strata is not associated with the production of the magnetic carrier. Remagnetization may be attributed to the alignment of the fine grains of hematite and pyrrhotite within calcium carbonate rich water in the interstices among quartz or calcite grains.

1. Introduction

The Korean peninsula occupies the eastern part of the stable South China Block. The fairly continuously formed sedimentary strata are cropped out in the peninsula (LEE, D. S., 1987). The strata cover in age about 600 Mys between the Cambrian and the Quaternary. Cenozoic and Mesozoic paleomagnetic data have been accumulated from the southern part of the peninsula in an attempt to establish the apparent polar wandering path through time to discuss the block motion of the South China Block as well as of the peninsula (ITO and TOKIEDA, 1980; OTOFUJI *et al.*, 1983, 1986; SHIBUYA *et al.*, 1985; LEE, G. *et al.*, 1987; LEE, D. S., 1987). Further paleomagnetic study of more ancient times is now required.

It has become apparent in the last few years that a large number of Mesozoic

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rocks of the Korean peninsula had a significant overprint in the present direction of earth's magnetic field (SHIBUYA *et al.*, 1985; OTOFUJI *et al.*, 1986; LEE, G. *et al.*, 1987). For paleomagnetic work this strongly requires that a careful stability test should be applied, and that the paleomagnetic study should be investigated in an area where a fold test is available.

We collected continental sediments from the Paegunsan Syncline in the central part of the Korean peninsula (Fig. 1), where a typical synclinal structure is observed. We originally anticipated that a Paleozoic to Mesozoic megnetostratigraphy could be constructed. We encountered, however, a typical remagnetized sedimentary sequence instead of a pre-Cenozoic paleomagnetic field direction. In this paper we present paleomagnetic and rock magnetic data of the remagnetized rocks of the Paegunsan Syncline, and discuss the remagnetization process.

2. Geological Setting and Sampling

The Paegunsan Syncline is located at latitude 37°10'N and longitude 128°50'E in the northeastern part of the Okch'on Fold Belt (Fig. 1). This syncline is elongated about 30 km east to west, with a relatively narrow width of 8–20 km. The Paegunsan Syncline has a hinge line of N30°E, and its synclinal axis plane dips gently towards north with the overturned north wing.

The syncline is covered by the Choson Supergroup (the Lower Paleozoic Sequence) and the P'yongan Supergroup (Permo-Carboniferous and Triassic Systems) (LEE, D. S., 1987). The two supergroups are most cropped out on the surface of this syncline in the Korean peninsula.

2.1 Choson Supergroup

This supergroup unconformably overlies Precambrian granite gneiss and metasediments (the Yulli Group), and is in turn unconformably overlain by the Carboniferous-Triassic strata of the P'yongan Supergroup. The Choson Supergroup has been classified from lithostratigraphic viewpoints into 10 units (LEE, H. Y., 1980) as shown in Fig. 2; the Changsan Quartzite, Myobong Slate, Taegi Formation, Sesong Slate, Hwajol Formation, Tongjom Quartzite, Tumudong Shale, Maktong Limestone, Chigunsan Shale and Tuwibong Limestone, in ascending order.

The Changsan quartizite, the basal unit of the Choson Supergroup, is overlain conformably by the Myobong Slate, in which fauna indicative of an early Cambrian age was discovered (YUN, S., 1978). The Cambrian age is assigned to the lowest part of the supergroup.

The upper part of the supergroup above the Myobong Slate is predominately carbonate rocks, called the Great Limestone Group. From the Taegi Formation to the Tuwibong Limestone, eight formations were conformably formed. Based on the conodonts, the Llandeilian to Carodocian of the Middle Ordovician is assigned to the uppermost part of the Choson Supergroup of the Tuwibong Limestone (LEE, H. Y., 1980).

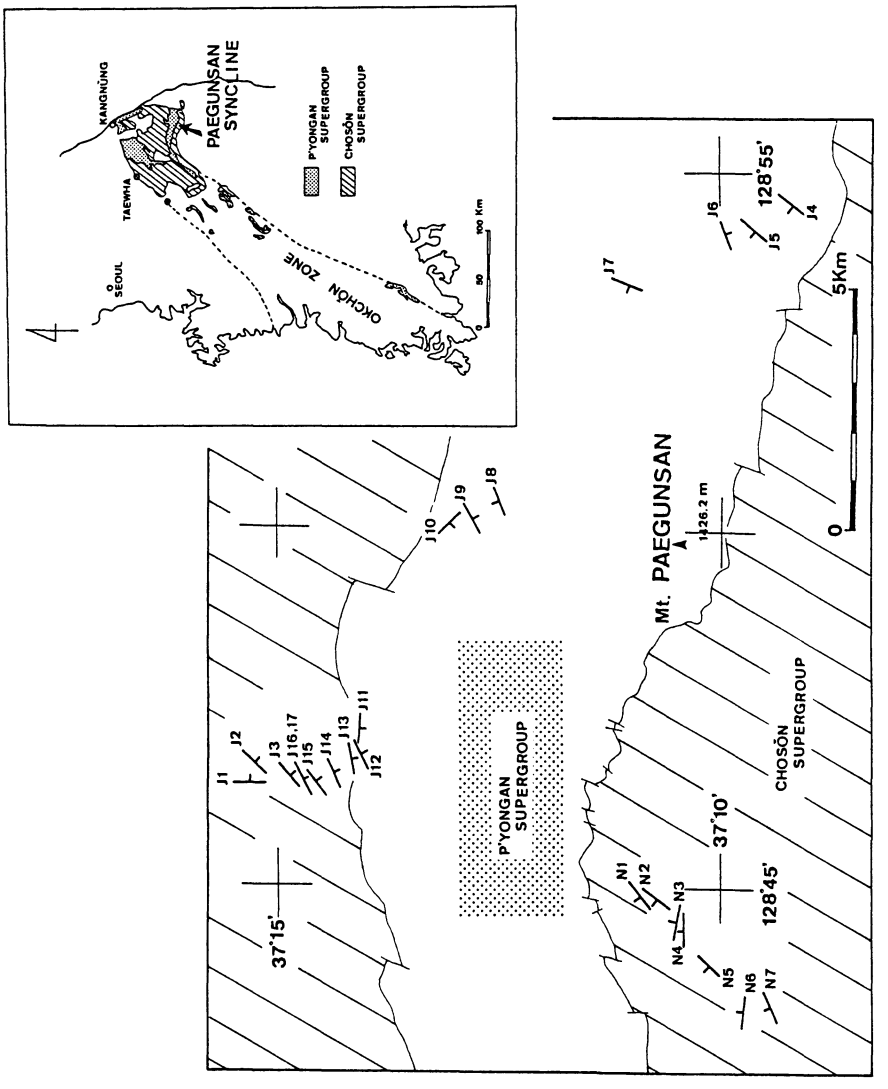


Fig. 1. Map of paleomagnetic sampling sites in the Paegunsan Syncline together with a simplified geologic map of south Korea, showing the distribution of the Choson and P'yongan Supergroup. Schematic strike and dip of the strata is drawn for each sampling site.

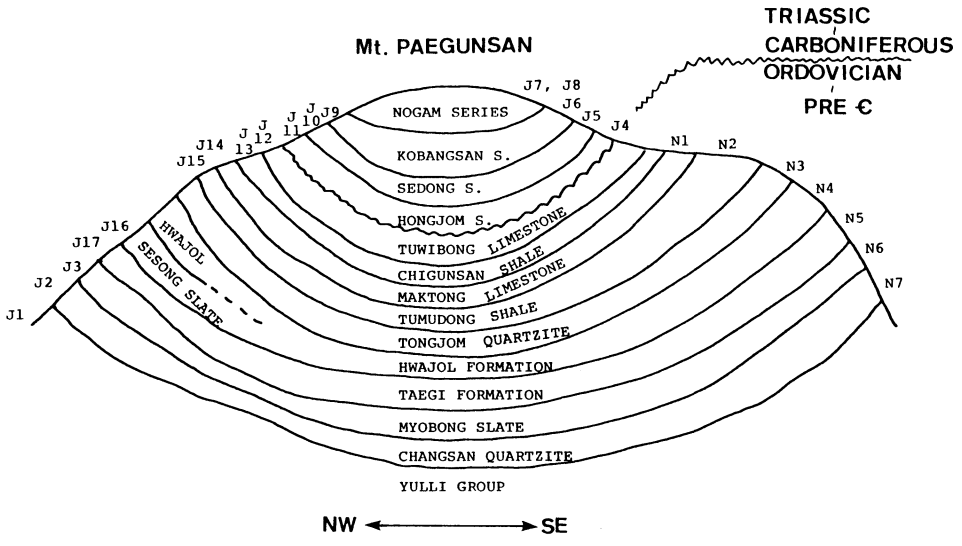


Fig. 2. Cross section through the Paegunsan Syncline and location of the paleomagnetic sampling sites.

2.2 *P'yongan Supergroup*

The P'yongan Supergroup disconformably overlies the Middle Ordovician limestone. The disconformity represents a great hiatus ranging from Late Ordovician to Early Carboniferous.

The supergroup is divided into 4 units on the basis of lithologic and paleontologic data (LEE, D. S., 1987); namely, the Hongjom Series, Sadong Series, Kobangsan Series and Nogam Series, in ascending order.

In contrast with the Great Limestone Group, the P'yongan Supergroup is characterized by shale and sandstone with variable colours. The lower part of the supergroup contains the intercalation of thin lenticular limestone beds. The upper part of the supergroup is composed of non-marine sands and muds. The top of the Nogam series is unconformably covered by the Chokkar Formation of Cretaceous age.

2.3 *Sampling*

Samples were collected at 24 sites along the two routes in the valleys of the northeastern and western sides of the foot of the Mt. Paegunsan (Fig. 1). These routes run nearly at right angles to the hinge line of the Paegunsan Syncline. Twelve in twenty-four sampling sites were on the north wing of the syncline, and the remaining sites were on the south wing. The samples covered the strata in age between the Precambrian (the Yulli Group) and Triassic (the Nogam Series) (Fig. 2).

Hand samples were taken and a magnetic compass was used for orientation. Each site typically comprised three independently oriented samples distributed over

distance ranging up to 10 m. The present declination value at each sampling point was not measured, but was calculated in the laboratory using the international geomagnetic reference field (IGRF) of 1984 (IAGA DIVISION I WORKING GROUP 1, 1987).

3. Paleomagnetic Study

In the laboratory, two or three specimens 25 mm in diameter and 25 mm long were prepared from the samples. No specimens were obtained for five sites because of the crushing of samples during drilling. The NRMs of the specimens were measured on a two-component ScT cryogenic magnetometer. Stability of the NRM was examined by both thermal and alternating field (AF) demagnetization. The thermal demagnetization was carried out with a laboratory built furnace which was set in three-fold mu-metal shield. The specimens were cooled after removal of the furnace from the shield, where residual magnetic field was less than 10 nT. The AF demagnetization of the specimens was treated with a three-axis tumbler system in a one cylindrical mu-metal shield. The preliminary demagnetization experiment for the pilot specimens showed little decay in NRM intensity during the AF demagnetization. All the remaining specimens were, therefore, progressively demagnetized in steps of 50°C using the thermal method.

The thermomagnetic analyses were done with a recording thermomagnetic balance on single specimens weighing between 70 and 200 mg. These specimens were heated in air in a magnetic field of 0.8 T. The heating and cooling rates were kept constant at 6.7°C/minute.

3.1 Measurements

Thermal demagnetization experiments revealed that the specimens had similar demagnetization behavior within each site. Sampling sites were divided into three categories according to the behavior during progressive thermal demagnetization of specimens: (A) specimens with unblocking temperatures of about 350°C, (B) specimens with unblocking temperatures of 670°C, and (C) specimens with no characteristic direction.

Category A (sites J2, J4, J5, J6, J7, J12, J13, J14, J15, J17, N1, N2)

Typical specimens of this category lost their NRMs, down to less than 3% in intensity of the original NRMs, at 350°C in thermal demagnetization, whereas more than 50% of NRMs remained up to 50 mT during AF demagnetization (Fig. 3). Unblocking temperatures were limited to narrow range between 300°C and 400°C. The NRMs of these specimens showed monotonous convergence to the origin in the orthogonal plot up to 300°C, and no characteristic direction was identified during further thermal demagnetization above 400°C (Fig. 3). Such demagnetization behavior indicates that specimens had only one magnetic component with an unblocking temperature of about 350°C.

The thermomagnetic curve showed an abrupt increase in magnetization above 400°C and the disappearance of magnetization at 580°C (Fig. 4), indicating that

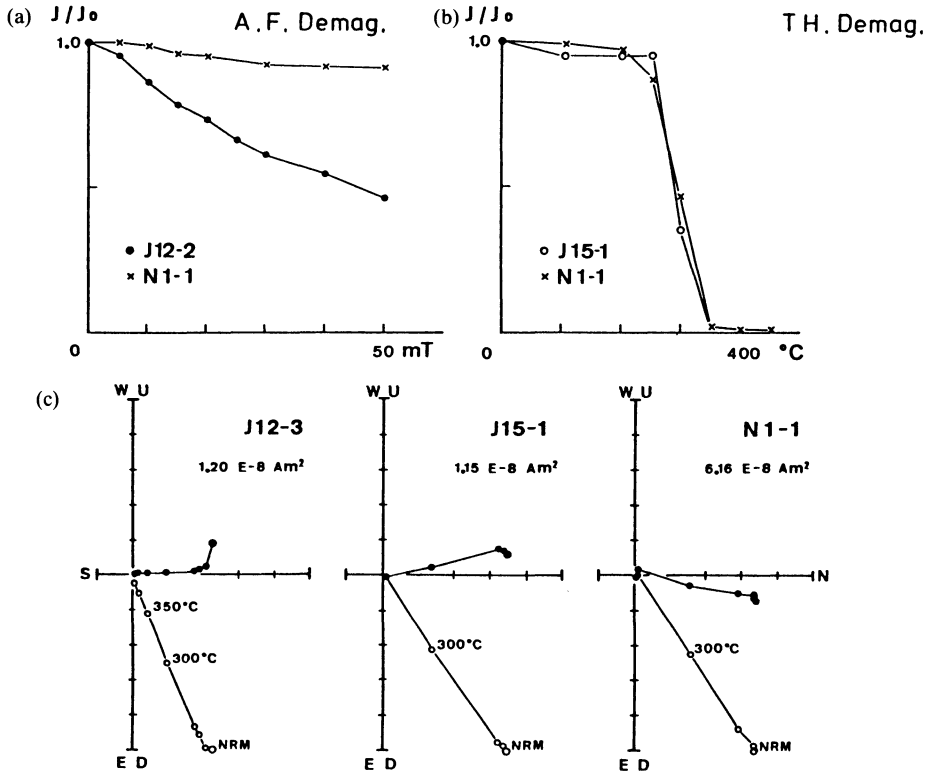


Fig. 3. Demagnetization behavior for specimens in the category A (see text). (a), (b) Normalized intensity decay during progressive alternating field and thermal demagnetization, respectively. (c) Vector thermal-demagnetization diagram. Open (solid) symbols show the magnetic vector projection on the vertical (horizontal) plane. Tectonic tilt corrections have not been applied.

some minerals were converted into magnetite. No magnetic minerals were identified from the thermomagnetic curve in the temperature range below 400° C.

Category B (sites J3, J16, N3, N6)

Hematite appeared to be the main carrier, and AF demagnetization was not effective. In almost all specimens of this category, thermal treatment revealed a single magnetic component with a demagnetization trajectory that converged toward the origin (Fig. 5). Unblocking temperatures of the magnetic component were between 670° C and 700° C, indicating that the magnetic carrier is hematite. The thermomagnetic curves (Fig. 4) support our conclusion, deduced from the thermal demagnetization, that the main carrier is hematite.

Two samples in site N3 showed a large directional change during demagnetization at a low temperature, 100° C: the directional change exceeded 80 degrees. The monotonous demagnetization was observed during further demagnetization processes. The NRM directions were variable in specimens of these samples before

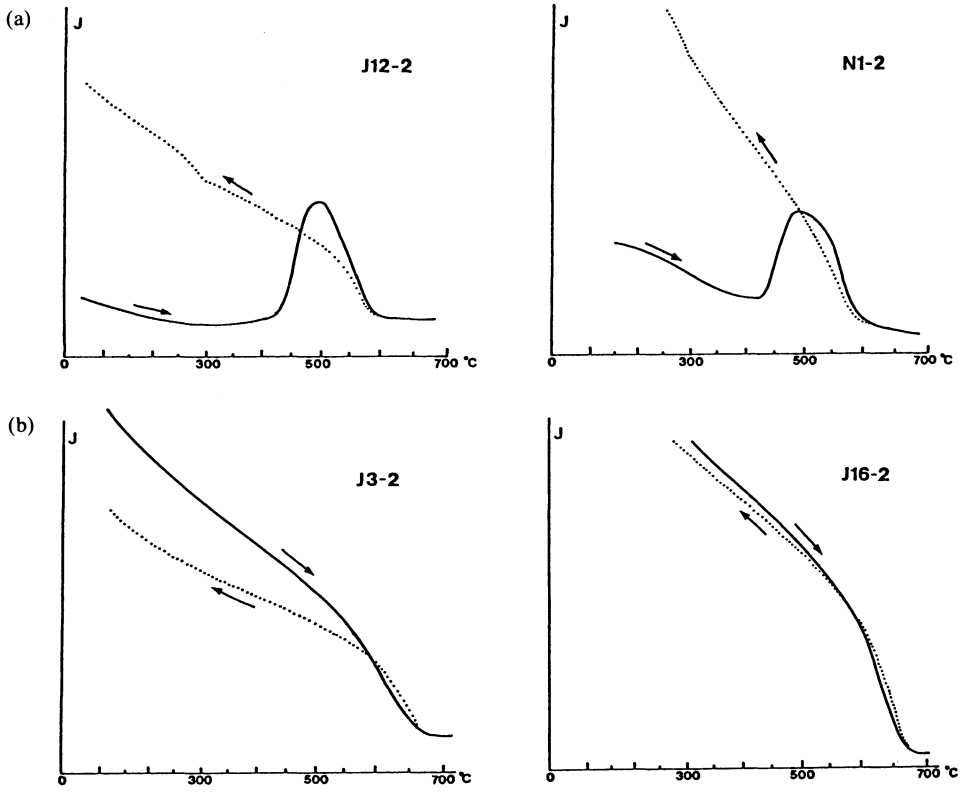


Fig. 4. Thermomagnetic curves in air in 0.8 T for specimens in the two categories A and B (see text). (a) Category A. (b) Category B. The intensity of magnetization is in arbitrary units. Arrows mark the heating and cooling cycle.

demagnetization. The low temperature component was probably acquired in the magnetic field during the storage in the laboratory.

Category C (sites J10, N7)

The thermal demagnetization diagrams displayed an erratic directional behavior. Their samples had weak NRMs: the intensity before demagnetization was less than 1×10^{-4} A/m. Specimens of this category hardly acquired stable NRM, which was in turn easily covered by the remanence induced in the laboratory magnetic field. These sites are not considered further in analysis.

3.2 Results

In 17 sites of category A and B, a characteristic direction of the specimens was isolated. The NRM direction after demagnetization at 200° C or 300° C was adopted as the characteristic direction for the specimens. Mean direction of each site was calculated by assigning a unit weight to a specimen. We arbitrarily discarded results

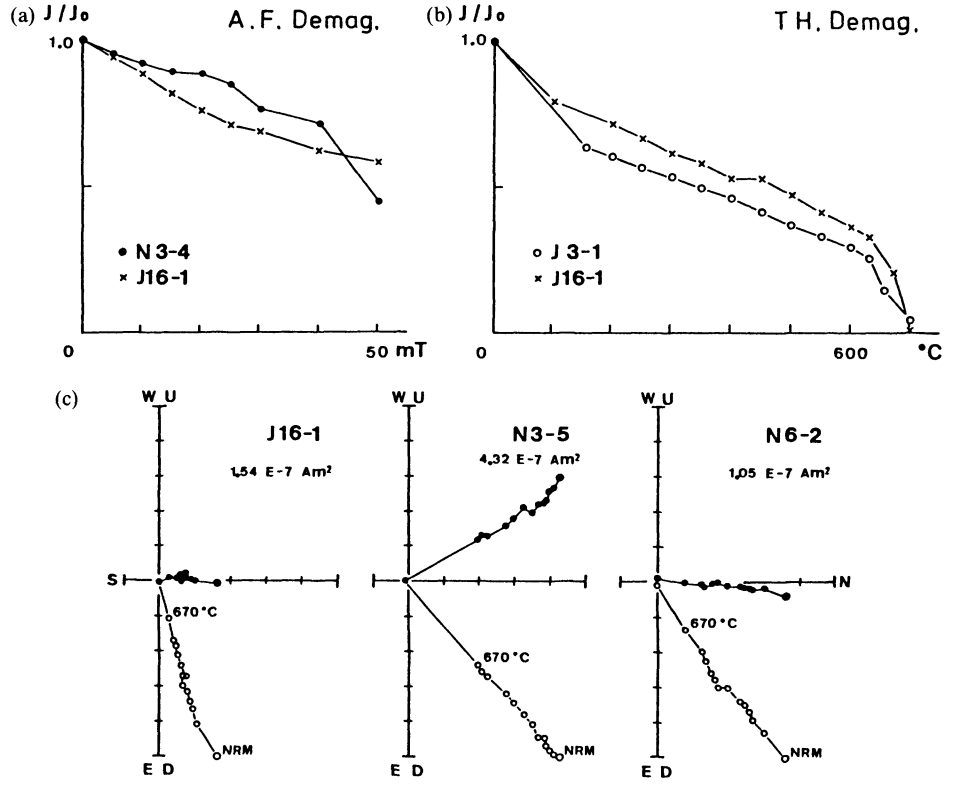


Fig. 5. Demagnetization behavior for specimens in the category B (see text). (a), (b) Normalized intensity decay during progressive alternating field and thermal demagnetization, respectively. (c) Vector thermal-demagnetization diagram. Open (solid) symbols show the magnetic vector projection on the vertical (horizontal) plane. Tectonic tilt corrections have not been applied.

from any sites for which the circle of 95% confidence exceeded 35°, to exclude poorly determined and spurious directions from the data set. Fifteen sites in 24 sites remained.

The NRM directions before and after tilt correction are shown in Table 1 and drawn in Fig. 6. All NRM directions in situ have a northerly direction with normal polarity. The NRM directions after tilt correction were distributed in two areas: a northwestward direction with a downward inclination and a southeastward one with a steeply downward inclination. The NRM directions in situ appear to be clustered more tightly than those after tilt correction.

Since we collected samples on either wing of the Paegunsan Syncline, the NRM directions from the north and south wings of the synclinal structure are compared (see Table 1). Samples of the Maktong limestone were collected at the sites of J14 and N1. The NRM directions from the two sites are more than 77.8 degrees apart

Table 1. Paleomagnetic results of the Paegunsan Syncline.

Formation name	Sampling site	Demag. (°C)		$N(n)$	D (°) (in situ)	I (°) (in situ)	D (°) (tilt correction)	I (°) (tilt correction)	k	α_{95} (°)
[P'YONGAN SUPERGROUP (Triassic~Carboniferous)]										
Nogam Series	J7	200	A	3(5)	21.6	57.0	−34.9	43.5	9.8	31.0
	J8									
Kobangsan S.	J6		A							35<
	J9									
Sadong S.	J5	200	A	3(6)	−18.2	41.6	−29.2	−2.7	5.1	33.0
	J10		C							
Hongjom S.	J4	200	A	3(4)	−2.1	46.8	−19.6	29.0	31.1	16.7
	J11									
[CHOSON SUPERGROUP (Ordovician~Cambrian)]										
Tuwibong Limestone	J12	200	A	3(5)	0.3	69.4	132.4	43.5	40.9	12.1
Chigunsan Shale	J13	200	A	3(5)	−0.4	62.3	143.5	65.5	172.3	5.8
Maktong Limestone	J14	200	A	3(5)	8.1	59.4	114.7	52.5	59.0	10.0
	N1	200	A	2(4)	9.8	52.6	−11.1	27.7	687.6	3.5
Tumudong Shale	J15	200	A	2(3)	−13.1	61.1	103.4	56.0	536.2	5.3
	N2	200	A	3(3)	11.3	60.2	−25.9	36.7	52.0	17.3
Tongjom Quartzite	N3	300	B	5(9)	−22.1	58.7	−12.0	30.3	46.6	7.6
Hwajol Formation	N4		B							35<
Sesong Slate	J16	300	B	3(6)	−8.0	68.8	114.6	70.6	72.8	7.9
Taegi Formation	J17	200	A	3(4)	17.4	69.2	94.3	52.6	225.0	6.1
	N5									
Myobong Slate	J3	300	B	3(5)	−13.6	54.6	−19.3	72.4	110.5	7.3
	N6	300	B	2(5)	−10.6	55.1	−7.5	23.3	173.5	5.8
Changsan Quartzite	J2	200	A	1(2)	−7.1	45.6	12.2	53.4	—	—
	N7		C							
(Pre Cambrian)										
Yulli Group	J1									

Paegunsan syncline	N	D (°)	I (°)	k	α_{95} (°)
in situ (before tilt correction)	15	−2.7	58.1	54.9	5.2
after tilt correction	15	8.3	61.7	3.5	23.9

Demagnetization: the magnetic cleaning temperature and the three categories (A,B and C) on the demagnetization behaviour (see text). No specimens could be obtained from samples for the sites J1, J8, J9, J11 and N5. $N(n)$: N is number of samples and n is number of specimens.
Mean direction of a site was calculated by assigning unit weight to specimens. D is the mean declination, I is the mean inclination, k is the Fisher precision parameter and α_{95} is the radius of the 95% circle of confidence around the site mean direction.

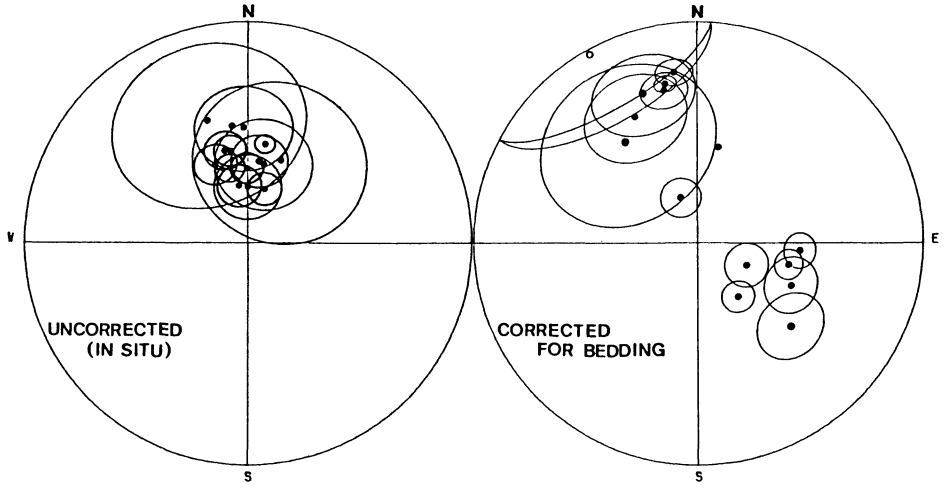


Fig. 6. Paleomagnetic field directions and 95% confidence circles of sites in the Paegunsan Syncline before and after correcting for bedding. Projections are equal area, solid symbols on the lower hemisphere, open symbols on the upper hemisphere.

after tilt correction, whereas the difference in the two directions in situ is only 6.9°. The NRM directions from the Tumudong shale and the Myobong slate also show little deviation between directions from either wing in situ. It is evident that the NRMs of these strata were acquired at some time after the strata had been folded.

Although the sedimentary strata of the Paegunsan Syncline represent such a extremely long time that the primary NRMs of these formations could have had variable directions, the apparent fold test is tentatively applied to a whole strata of the Paegunsan Syncline (Table 1). The NRM directions in situ are tightly clustered and had a large precision parameter (K=54.9), whereas those after tilt correction are distributed along a small circle plane and show a small precision parameter (K=3.5). The fold test is clearly negative at the 99% probability level (MC ELHINNY, 1964). The sedimentary strata in the Paegunsan Syncline are presumed to have lost their primary NRM and acquired their secondary NRMs after the synclinal structure had been built.

4. Remagnetization

The main result that emerges from this study is that only one direction can be identified in the NRMs of the Choson and P'yongan Supergroups of the Paegunsan Syncline, and that it was acquired at some time after the folding occurred in the Paegunsan region. Although we fail to obtain the paleo-geomagnetic field directional change from the Cambrian to the Triassic, we discover a typical, remagnet-

ized, sedimentary rock sequence in the Paegunsan Syncline.

4.1 Timing of the remagnetization

The mean NRM direction of the supergroups in situ ($D=-2.7°$, $I=58.1°$, $\alpha_{95}=5.2°$; see Table 1) is very close to the present axial dipole field ($D=0°$, $I=57°$)(Fig. 7): the deviation is only 2.1°. The mean NRM direction deviates by 6.3° from the present geomagnetic field. This deviation exceeds the amount of the 95% confidence circle of 5.2° for the mean NRM direction. The mean NRM direction is compared with the Cretaceous paleomagnetic direction ($D=29.3°$, $I=59.4°$) expected from the Kyongsang Basin of south Korea (OTOFUJI *et al.*, 1986). The deviation between the two directions is 16.5 degrees. These large deviations require a tectonic movement so that the Paegunsan Syncline dipped toward the south by at least 31° after the supergroup had acquired the secondary NRM in the Cretaceous. No geological study of the Paegunsan Syncline supports this requirement (LEE, D. S., 1987). The expected direction of the period older than the Cretaceous can be calculated from the apparent polar wandering path of the South China Block (LIN *et al.*, 1985). More than 13 degrees in deviation is observed between the observed mean NRM direction and the expected ones from the periods older than the Cretaceous. On the basis of these comparisons, it is reasonable that the observed mean NRM direction is parallel to the axial dipole field and that the remagnetization occurred quite recently, in the Brunhes epoch. The duration of the remagnetization is longer than the period in which directional change due to geomagnetic field secular variation is averaged out.

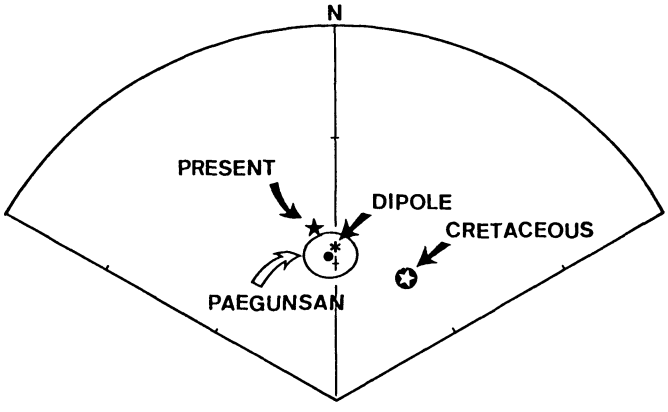


Fig. 7. Comparison of field direction at the Paegunsan Syncline (37°10′N, 128°50′E). Closed circle with 95% confidence circle: Paleomagnetic field direction of the Paegunsan Syncline before bedding correction. Closed star: present field direction ($D=-7.4°$, $I=52.4°$) calculated from IGRF 1984 (IAGA DIVISION I WORKING GROUP 1, 1987). Asterisk: present axial dipole field direction ($D=0°$, $I=56.6°$). Open star: Cretaceous paleomagnetic direction recalculated from the date of the Kyongsang Basin ($D=29.3°$, $I=59.4°$; OTOFUJI *et al.*, 1986; $D=28.0°$, $I=58.5°$; LEE *et al.*, 1987).

4.2 Magnetic carrier

The formations in the Paegunsan Syncline are divided into two categories according to unblocking temperature during thermal demagnetization: 670°C and 350°C. Three formations (the Myobong, Sesong, and Tongjom) show the high unblocking temperature of about 670°C, while the others show 350°C. This implies that two magnetic minerals play an important part in remagnetization of the Paegunsan Syncline.

Hematite is a prominent candidate for the magnetic carrier with the high unblocking temperature. The thermomagnetic curve strongly indicates hematite as one of the principal magnetic minerals of remagnetization (Fig. 4). The magnetization disappeared at about 670°C. Nearly reversible behavior was observed during thermomagnetic analysis in a sample from site J16, while the thermomagnetic curve was irreversible in the sample from site N3. A small fraction of titanomagnetite is responsible for this irreversible behavior. The microscopic study supports this conclusion: hematite is abundant in these sites. Hematite grains are in particular distributed between the grain boundary of quartz or calcite (Fig. 8). Titanomagnetite with illmenite lamellae is observed in site N3 but is of rare occurrence.

Pyrrhotite is the principal magnetic mineral in sedimentary strata with a low unblocking temperature. The unblocking temperature between 300°C and 350°C (Fig. 9) is compatible with the blocking temperature of polycrystalline pyrrhotite with the Curie temperature of 325°C (HALGEDAHL and FULLER, 1981). Microscopic study shows that sulfide minerals (e.g., pyrite) are abundant. The pyrrhotite occurs at a margin of the euhedral pyrite (Fig. 8). Undetermined sulfide minerals (pyrite and/or pyrrhotite) fill the interstices between quartz grains or carbonate crystals. Hematite, titanomagnetite and titanomaghemite are not identified under the microscopic observation. The magnetization is too weak in thermomagnetic analysis during the low temperature range between room temperature and 400°C to identify the Curie temperature of pyrrhotite (Fig. 4). The increase in the magnetic moment at about 400°C is probably due to chemical change from pyrite and pyrrhotite to magnetite during heating.

The NRM directions of formations carried by hematite and pyrrhotite are compared in situ. The two NRMs have the same direction within 95% confidence limit (hematite: $D=-13.9°$, $I=59.4°$, $\alpha_{95}=8.2°$ vs. pyrrhotite: $D=1.2°$, $I=57.4°$, $\alpha_{95}=6.5°$). The sedimentary strata with different magnetic carriers recently acquired their new NRMs at the same time.

4.3 Origin of hematite and pyrrhotite

Hematite is contained in the Myobong Slate, the Sesong Slate and the Tongjom Quartzite. Pyrrhotite is in the limestone, sand stone and shale of other formations. Two possibilities arise for origin of these minerals: (1) diagenesis and (2) metamorphism.

Chemical interaction in accumulated sediments can produce authigenic pyrrhotite and hematite. The iron ions are produced from iron bearing minerals, such as olivines, pyroxens and amphiboles in sediments, and further form iron

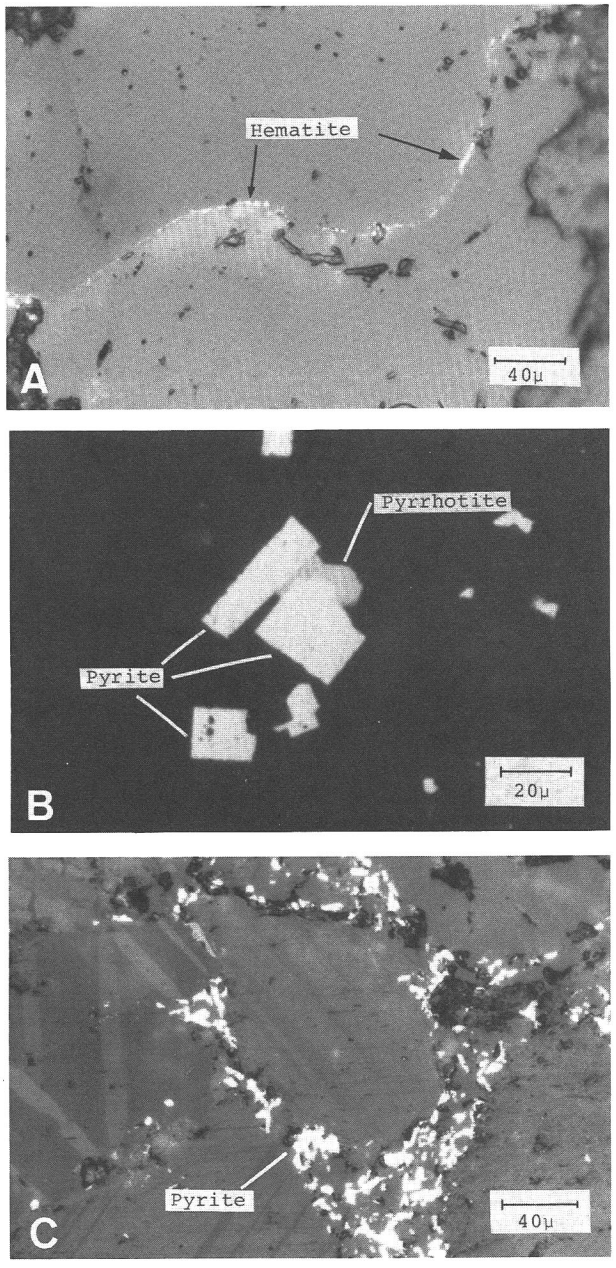


Fig. 8. Polished surface photomicrograph, vertically reflected illumination. (A) Hematite is distributed in the interstice between quartz grains. Specimen of N3-2 in the category B (see text). (B) Pyrrhotite and pyrite. Pyrrhotite occurs between euhedral pyrite. Specimen of N1-2 in the category A (see text). (C) Pyrite fills cleavage among calcium carbonate crystals. Specimen of J14-1 in the category A (see text).

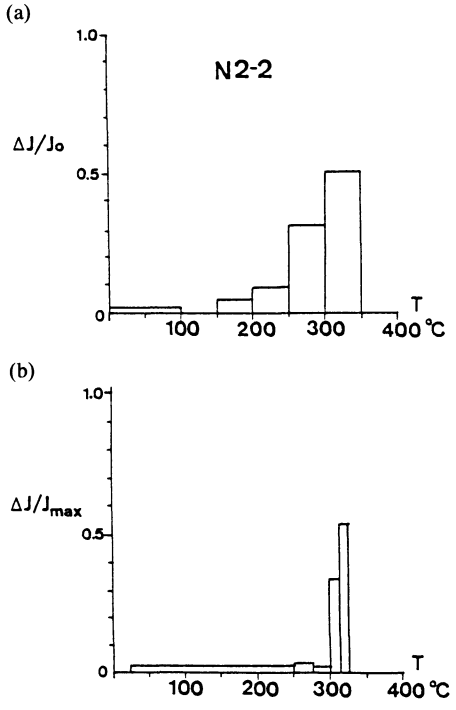


Fig. 9. Comparison of unblocking and blocking temperature spectra of remanent magnetization. (a) The unblocking temperature spectrum of NRM of specimens (N2-2) in the category A (see text) of the Paegunsan Syncline. (b) The blocking temperature spectrum of PTRM of polycrystalline pyrrhotite with Curie temperature of 325°C (HALGEDAHL and FULLER, 1981). $\Delta J = |J(T_{j+1}, T_R) - J(T_j, T_R)|$ when T_R=Room temperature. J_0=NRM of the specimen before thermal demagnetization. J_{max}=total TRM carried by the pyrrhotite sample.

hydroxides or oxides. When they encounter any hydrogen sulfide being released by microbial activity, pyrite is formed. Pyrrhotite instead of pyrite forms in conditions of very high pH and low pE (HENSHAW and MERRILL, 1980). It takes some 500 years of deposition for pyritization to take place (MANNING *et al.*, 1979). Unless the sediments contain a source of hydrogen sulfide, the iron hydroxides and oxide are likely to be oxidized to hematite, which takes a few million years.

Metamorphism is the other factor in the production of magnetic minerals. When rocks bearing, Fe, TiO_2 and S_2 are heated, pyrrhotite or hematite can be formed (MIYASHIRO, 1965). The alternative production of pyrrhotite or hematite is controlled by the O_2 and S_2 fugacities (ITAYA *et al.*, 1985). Pyrrhotite never coexists with hematite, whereas magnetite can coexist with hematite. Metamorphism can produce a sedimentary layer which bears either hematite or pyrrhotite.

The two hypotheses were examined through microscopic observation. Micro-

scopic study indicates that hematite-bearing slate and quartzite contain much authigenic magnesian and ferric minerals. They are composed of chlorite, white mica, and biotite. Authigenic fine magnesian and ferric minerals are also observed in the pyrrhotite-bearing Tumudong limestone. These facts strongly indicate that the sedimentary layers in the Paegunsan Syncline were more or less metamorphosed. We conclude that pyrrhotite and hematite are due to metamorphism: the hematite and pyrrhotite were probably produced at the same time when chlorite, mica and biotite occurred in conditions above 400°C, at intermediate pressure.

4.4 Timing of the production of magnetic minerals

The Paegunsan region was subjected to tectonic disturbance several times (LEE, D. S., 1987). The preliminary movement is attributed to the Triassic Songnim Disturbance, and then a large scale tectonic movement associated with the Daebo Disturbance followed in the Jurassic time. Intermediate to felsic Bulguksa Granites intruded in the Cretaceous period, and this plutonic activity lasted until the Paleogene. The Paegunsan Syncline has had many chances to be subjected to metamorphism.

The Okch'on Metamorphism Belt in the Okch'on Zone is located immediately to the southwest of the Paegunsan Syncline. This belt is an intermediate-pressure type of regional metamorphism development (KIM, 1971; LEE, S. M., 1972). The Taebaeksan basin including the Paegunsan Syncline underwent low grade metamorphism. The age of the metamorphism of the Okch'on Zone has been expected to be at the end of the Paleozoic or the beginning of the Mesozoic (HIROI, 1981: LEE, D. S., 1987).

Our paleomagnetic study indicates, however, that the remagnetization occurred in recent years, probably during the Brunhes epoch. This implies that the magnetic carriers of hematite and pyrrhotite may have been produced during the last 0.7 Mys in the Brunhes epoch, if the remagnetization took place accompanying the appearance of the magnetic carriers. This implication is ridiculous, because we have no geological evidence that the Paegunsan Syncline was subjected to an extremely violent disturbance within the Brunhes epoch (LEE, D. S., 1987). The sediments of the Paegunsan Syncline have been buried during a period of more than 200 Mys. The probability is quite low, less than 0.4% (0.7 Mys/200 Mys), for the sediments in the Paegunsan Syncline to have been metamorphosed within the last 0.7 Mys in the Brunhes epoch. The P'yongan and Choson Supergroups were probably metamorphosed at the same time as the Okch'on Metamorphism Belt underwent metamorphism. We conclude that the magnetic carriers hematite and pyrrhotite were produced before the Brunhes epoch.

5. Conclusion and Speculation

The nature of the NRMs of the Paegunsan Syncline is summarized as follows.

(1) The sedimentary rocks were subjected to remagnetization after the synclinal structure had been built.

(2) The remagnetization process worked during the last 0.7 Mys in the Brunhes Epoch.

(3) The NRMs of remagnetization are carried by either hematite or pyrrhotite.

(4) Hematite and pyrrhotite occurred in the beginning of the Mesozoic when regional metamorphism developed in the Okch'on Zone.

This summary strongly indicates that the origin of the NRMs of the Paegunsan Syncline is not ascribed to chemical remanent magnetization which arose associated with the generation of hematite and pyrrhotite during metamorphism.

Remagnetization acquired in the recent period is rather explained by viscous partial thermo remanent magnetization (VTRM) (PULLAIAH *et al.*, 1975; KENT, 1985) and/or NRM due to weathering. The former is an inevitable factor in recent remagnetization. We stress the role of the weathering, because we find some evidence for weathering which is responsible for the remagnetization of sedimentary rocks of the Paegunsan Syncline.

Microscopic study shows that quartz and calcite grains are now dissolved and decreased in volume, indicating that the grain boundary becomes a permeable space for pore water. Some pyrite is transformed to goethite. This suggests that weathering has progressed more or less into the interior of the sedimentary rocks in the Paegunsan Syncline. The grain boundary of the quartz and calcite is filled with opaque minerals (e.g., hematite or pyrite) (Fig. 8). Magnetic carriers have quite small grain sizes, because little remanence decreases during the AF demagnetization up to 50 mT on samples with either pyrrhotite or hematite. These fine magnetic particles are small enough to move and rotate within the permeable interstice space among the quartz and calcite grains.

Since porosity generally develops by outcrop dissolution of calcite cement among grains during weathering (MCBRIDE, 1987), sedimentary rocks are likely to be permeable after weathering. The new NRM can originate from alignment of magnetic grains of hematite or pyrrhotite to the ambient magnetic field within pore water containing calcium carbonate with high viscosity among quartz or carbonate grains. This magnetization process is analogous to that of the post-DRM (CREER *et al.*, 1983). The high viscosity of the pore water is responsible for the long relaxation time, about tens of thousands years.

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